Daniel Stilck França

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Research Interests

Quantum algorithms, quantum learning and information theory; mathematical physics and optimization.

Academic Appointments

09.2024–present **Associate Professor**, University of Copenhagen.
Quantum for Life, Department of Mathematical Sciences.

04.2022–08.2024 **Inria Starting Faculty Position**, École Normale Supérieure de Lyon.
LIP, Department of Computer Science; QInfo team. Currently on leave from Inria.

09.2018–03.2022 **Postdoctoral Researcher**, University of Copenhagen.
QMATH, Department of Mathematical Sciences. Supervisor: Matthias Christandl.

04.2013–07.2018 **Teaching Assistant**, Technical University of Munich.
Mathematical Physics Department.

Education

04.2015–09.2018 **Dr. rer. nat.**, Technical University of Munich.
Thesis: *Irreversibility in Quantum Information Theory.*
Supervisor: Michael M. Wolf.

10.2014–03.2015 **M.Sc. in Mathematics, with honors**, Technical University of Munich.

04.2011–09.2013 **B.Sc., with honors**, Technical University of Munich.
Major: Mathematics; minor: Physics.

Honors and Scholarships

2018 **Reviewer of 2018**, *Journal of Physics A: Mathematical and Theoretical.*

2016 **Hurwitz Society Prize** for an exceptional master's thesis.

2011–2016 **DAAD Scholarship**, German Academic Exchange Service; full scholarship.

Research and Education Funding

2027–2028 **Quantum Readiness for Innovation (QRI)**. DeIC quantum-competence grant. Award: **DKK 3.0 million**. Course lead for the MSc Quantum Innovation course.

2026 **PROQSY**. Horizon Europe MSCA European Postdoctoral Fellowship. Award: **EUR 263,000**. Grant holder.

2025–2030 **GIFNEQ: Gibbs framework for near-term quantum computing**. ERC Starting Grant. Award: **EUR 1.5 million**. Principal investigator. Project period: 1 July 2025–30 June 2030.

2025 **PhD Scholarship Quantum Algorithms**. Danish e-Infrastructure Cooperation (DeIC). Award: **EUR 352,000**. Grant holder.

2025 **Training for Large-Scale Simulations of Quantum Systems**. DeIC Q-Competence grant. Consortium award: **EUR 507,000**. Consortium member; principal investigator: Mark Kamper Svendsen.

2023–2027 **TouQan: Towards a useful quantum advantage**. QuantERA 2023. Funding: **EUR 300,000**.

2023–2027 **Hybrid HPC Quantum Initiative**. French PEPR programme. Research-group funding: **EUR 336,000**. Coordinator of a EUR 3 million work package.

2022–2025 **Equality: Efficient Quantum Algorithms for Industry**. Horizon Europe. Funding: **EUR 250,000**.

Research Group and Supervision

Doctoral Researchers

- **Yufei Wang**, PhD student.
- **Tim Madsen**, PhD student; co-supervised with Rolf Poulsen.
- **Enrique Escobar Fernandez-Marcote**, PhD student through the QIS Graduate School; co-supervised with Jonas S. Neergaard-Nielsen and Marco Fanizza.

Completed Doctoral Supervision

- **Emily Beatty** (2022–2026, ENS Lyon); co-supervised with Guillaume Aubrun. Subsequently software developer at Bending Spoons. Best poster award, Beyond IID 2023, for joint work.
- **Victor Martinez** (2023–2026, ENS Lyon); co-supervised with Omar Fawzi. Subsequently permanent researcher at IBM Quantum Zurich.

Research Group and Supervision (continued)

Postdoctoral Researchers

- **Zhong-Xia Shang**.
- **Rafael A. Macêdo** (since May 2026).
- **Roberto Rubboli** (since October 2025).
- **Sam Slezak** (since October 2025); co-supervised with Mischa Woods.
- **Francisco Escudero Gutierrez**; co-supervised with Cambyse Rouzé.

Former Postdoctoral Researchers

- **Ngoc Hoang Anh Mai** (November 2023–October 2025). Subsequently permanent researcher at VAST, Hanoi, Vietnam.

Teaching and Curriculum Development

2026–2027	**Quantum Innovation**, University of Copenhagen. MSc course lead, supported by the QRI DeIC grant.
2026–2027	**Large-Scale Simulations of Quantum Systems**, University of Copenhagen. MSc module lead: quantum simulation on quantum computers.
Fall 2026	**Advanced Vector Spaces**, University of Copenhagen. MSc course lead, 7.5 ECTS. Linear and multilinear algebra, tensor products, spectral theory and Hilbert spaces.
2022–2024	**Quantum Computer Science**, ENS Lyon. Lecturer in academic years 2022–2023 and 2023–2024.
2020, 2021	**Introduction to Quantum Computing**, University of Copenhagen. Lecturer.
2019	**Introduction to LaTeX**, University of Copenhagen. Lecturer.
2013–2018	**Teaching assistant**, Technical University of Munich. Analysis 1–4 for Physicists, Mathematical Modelling and Markov Chains.

Industry Research Experience

03.2022–08.2023	**Research Consultant**, Zapata Quantum Computing. Development of quantum algorithms with reduced quantum-hardware requirements.

Languages

German and Portuguese (native); English (full professional proficiency); French (intermediate).

Academic Service

Conference Committees

2024 **Program committee chair**, QCTIP.

2020–2026 **Program committee member**: QIP (2024, 2026); TQC (2023, 2025); QCTIP (2022, 2026); Beyond IID (2022, 2025); QTURN (2020).

Conference and Workshop Organization

March 2024 **Main organizer**, Workshop on Fundamental Limitations to Quantum Computation, Banff, Canada.

June 2023 **Co-organizer**, Lyon-Singapore Quantum Information Meeting.

July 2021 **Organizer**, SIAM AG 21 Minisymposia on Tensor Networks and Geometry.

Journal Refereeing

Annales Henri Poincaré; *Communications in Mathematical Physics*; *IEEE Transactions on Information Theory*; *Journal of Mathematical Physics*; *Journal of Physics A: Mathematical and Theoretical*; *PRX Quantum*; *Quantum*; *ACM Transactions on Quantum Computing*; *Physical Review Letters*; *Nature*.

Outreach and Public Engagement

Selected Media Coverage

- *Estadão*: profile on quantum computing (2025, in Portuguese); named a "Brazilian to watch out for."
- *Inria*: ERC Starting Grant announcement.
- *Phys.org*: How noise limits today's quantum circuits.
- *Phys.org*: coverage of research on quantum error mitigation.
- *Physics World*: Conquering the challenge of quantum optimization.
- *Quanta Magazine*: Computer Scientists Prove That Heat Destroys Quantum Entanglement.
- *Nature Computational Science*: Analyzing noise for quantum advantage.

Public Talks and Interviews

Talks and round tables on quantum technologies for the Danish business and scientific community. Monthly interviews on Canal Resistentes (in Portuguese), since March 2020.

- 2023-02 *Limitations of noisy quantum devices*, Lyon Informatics Federation, **ENS Lyon**, Outreach talk, Lyon, France.
- 2021-12 *Steps towards a universal quantum computer*, Digital Tech Summit 2021, **Oksenhallen Copenhagen**, Panel discussion, Copenhagen, Denmark.
- 2021-05 *Quantum and big data*, Business Seminar, **Federal University of Rio Grande do Sul**, Panel discussion, Porto Alegre, Brazil.

Publications

Google Scholar | ORCID | arXiv

Journal Articles (Published or Accepted)

J1. T. Kraft, M. K. Joshi, W. T. Lam, T. Olsacher, F. Kranzl, J. Franke, L. Kh. Joshi, R. Blatt, A. Smerzi, D. Stilck França, B. Vermersch, B. Kraus, C. F. Roos, P. Zoller.
Bounded-error quantum simulation via Hamiltonian and Lindbladian learning
Physical Review X, accepted 1 July 2026.

J2. V. Martinez, O. Fawzi, D. Stilck França.
Sampling (noisy) quantum circuits through randomized rounding
Quantum, 10:2068, (2026).

J3. A. A. Mele, A. Angrisani, S. Ghosh, S. Khatri, J. Eisert, D. Stilck França, Y. Quek.
Noise-induced shallow circuits and the absence of barren plateaus
Nature Physics, 1–6, (2026).

J4. E. Beatty, D. Stilck França.
Order-p quantum Wasserstein distances from couplings
Annales Henri Poincaré, 27(3):787–845, (2026).

J5. C. Rouzé, D. Stilck França, Á. M. Alhambra.
Optimal quantum algorithm for Gibbs state preparation
Physical Review Letters, 136(6):060601, (2026).

J6. C. Rouzé, D. Stilck França, Á. M. Alhambra.
Efficient thermalization and universal quantum computing with quantum Gibbs samplers
Nature Physics, published online 15 April 2026, (2026).

J7. R. Ibarrondo, D. Stilck França.
Average contraction coefficients of quantum channels
Communications in Mathematical Physics, accepted, (2026).

J8. M. Fanizza, C. Rouzé, D. Stilck França.
Efficient Hamiltonian, structure and trace distance learning of Gaussian states
Nature Communications, accepted, (2026).

J9. J. J. Meyer, S. Khatri, D. Stilck França, J. Eisert, P. Faist.
Quantum metrology in the finite-sample regime
PRX Quantum, 6(3):030336, (2025).

J10. V. Martinez, A. Angrisani, E. Pankovets, O. Fawzi, D. Stilck França.
Efficient simulation of parametrized quantum circuits under nonunital noise through Pauli backpropagation
Physical Review Letters, 134(25):250602, (2025).

J11. O. Fawzi, A. Oufkir, D. Stilck França.
Lower bounds on learning Pauli channels with individual measurements
IEEE Transactions on Information Theory, 71(4):2642–2661, (2025).

J12. G. Wang, D. Stilck França, G. Rendon, P. D. Johnson.
Efficient ground-state-energy estimation and certification on early fault-tolerant quantum computers
Physical Review A, 111(1):012426, (2025).

J13. Y. Quek, D. Stilck França, S. Khatri, J. J. Meyer, J. Eisert.
Exponentially tighter bounds on limitations of quantum error mitigation
Nature Physics, 20(10):1648–1658, (2024).

J14. C. Rouzé, D. Stilck França, E. Onorati, J. D. Watson.
Efficient learning of ground and thermal states within phases of matter
Nature Communications, 15:7755, (2024).

J15. D. Harley, I. Datta, F. Ravn Klausen, A. Bluhm, D. Stilck França, A. H. Werner, M. Christandl.
Going beyond gadgets: the importance of scalability for analogue quantum simulators
Nature Communications, 15:6527, (2024).

J16. C. Rouzé, D. Stilck França.
Learning quantum many-body systems from a few copies
Quantum, 8:1319, (2024).

J17. D. Stilck França, L. A. Markovich, V. V. Dobrovitski, A. H. Werner, J. Borregaard.
Efficient and robust estimation of many-qubit Hamiltonians
Nature Communications, 15:311, (2024).

J18. G. Wang, D. Stilck França, R. Zhang, S. Zhu, P. D. Johnson.
Quantum algorithm for ground state energy estimation using circuit depth with exponentially improved dependence on precision
Quantum, 7:1167, (2023).

J19. C. Hirche, C. Rouzé, D. Stilck França.
Quantum differential privacy: An information theory perspective
IEEE Transactions on Information Theory, 69(9):5771–5787, (2023).

J20. G. De Palma, M. Marvian, C. Rouzé, D. Stilck França.
Limitations of variational quantum algorithms: a quantum optimal transport approach
PRX Quantum, 4(1):010309, (2023).

J21. C. Hirche, C. Rouzé, D. Stilck França.
On contraction coefficients, partial orders and approximation of capacities for quantum channels
Quantum, 6:862, (2022).

J22. D. Stilck França, R. García-Patrón.
A game of quantum advantage: linking verification and simulation
Quantum, 6:753, (2022).

J23. F. G. S. L. Brandão, R. Kueng, D. Stilck França.
Faster quantum and classical SDP approximations for quadratic binary optimization
Quantum, 6:625, (2022).

J24. D. Stilck França, R. García-Patrón.
Limitations of optimization algorithms on noisy quantum devices
Nature Physics, 17(11):1221–1227, (2021).

J25. D. Stilck França, S. Strelchuk, M. Studziński.
Efficient classical simulation and benchmarking of quantum processes in the Weyl basis
Physical Review Letters, 126(21):210502, (2021).

J26. M. Christandl, F. Gesmundo, D. Stilck França, A. H. Werner.
Optimization at the boundary of the tensor network variety
Physical Review B, 103(19):195139, (2021).

J27. I. Bardet, M. Junge, N. Laracuente, C. Rouzé, D. Stilck França.
Group transference techniques for the estimation of the decoherence times and capacities of quantum Markov semigroups
IEEE Transactions on Information Theory, 67(5):2878–2909, (2021).

J28. J. Borregaard, M. Christandl, D. Stilck França.
Noise-robust exploration of many-body quantum states on near-term quantum devices
npj Quantum Information, 7:45, (2021).

J29. A. Bluhm, M. Christandl, F. Gesmundo, F. Ravn Klausen, L. Mančinska, V. Steffan, D. Stilck França, A. H. Werner.
SARS-CoV-2 transmission routes from genetic data: A Danish case study
PLOS ONE, 15(10):e0241405, (2020).

J30. E. P. Hanson, C. Rouzé, D. Stilck França.
Eventually entanglement breaking Markovian dynamics: Structure and characteristic times
Annales Henri Poincaré, 21(5):1517–1571, (2020).

J31. A. Bluhm, D. Stilck França.
Dimensionality reduction of SDPs through sketching
Linear Algebra and its Applications, 563:461–475, (2019).

J32. D. Stilck França, A. K. Hashagen.
Approximate randomized benchmarking for finite groups
Journal of Physics A: Mathematical and Theoretical, 51(39):395302, (2018).

J33. D. Stilck França.
Perfect sampling for quantum Gibbs states
Quantum Information and Computation, 18(5–6), (2018).

J34. A. Müller-Hermes, D. Stilck França.
Sandwiched Rényi convergence for quantum evolutions
Quantum, 2:55, (2018).

J35. A. Müller-Hermes, D. Stilck França, M. M. Wolf.
Entropy production of doubly stochastic quantum channels
Journal of Mathematical Physics, 57(2):022203, (2016).

J36. A. Müller-Hermes, D. Stilck França, M. M. Wolf.
Relative entropy convergence for depolarizing channels
Journal of Mathematical Physics, 57(2):022202, (2016).

Peer-Reviewed Conference Proceedings

C1. C. Rouzé, D. Stilck França, Á. M. Alhambra.
Efficient thermalization and universal quantum computing with quantum Gibbs samplers
Proceedings of the 57th Annual ACM Symposium on Theory of Computing (STOC), 1488–1495, (2025).

C2. M. C. Caro, T. Gur, C. Rouzé, D. Stilck França, S. Subramanian.
Information-theoretic generalization bounds for learning from quantum data
Proceedings of the 37th Conference on Learning Theory (COLT), 775–839, (2024).

C3. D. Stilck França, F. G. S. L. Brandão, R. Kueng.
Fast and robust quantum state tomography from few basis measurements
Proceedings of the 16th Conference on the Theory of Quantum Computation, Communication and Cryptography (TQC), LIPIcs 197, 7:1–7:13, (2021).

Preprints

P1. Z.-X. Shang, D. Stilck França.
Hamiltonian dynamics from pure dissipation
arXiv preprint arXiv:2604.18533, (2026).

P2. R. T. Birke, J. B. Severin, M. A. Marciniak, E. Hogedal, A. Nylander, I. Ahmad, et al.
Demonstrating and benchmarking classical shadows for Lindblad tomography
arXiv preprint arXiv:2602.14694, (2026).

P3. S. Slezak, M. Scandi, Á. M. Alhambra, D. S. França, C. Rouzé.
Polynomial-time thermalization and Gibbs sampling from system-bath couplings
arXiv preprint arXiv:2601.16154, (2026).

P4. D. Stilck França, N. H. A. Mai.
Quantum speed-ups for solving semidefinite relaxations of polynomial optimization
arXiv preprint arXiv:2511.14389, (2025).

P5. H. Yuan, D. Stilck França, I. Luchnikov, E. Tiunov, T. Haug, L. Aolita.
Exponential speed-ups for structured Goemans–Williamson relaxations via quantum Gibbs states and Pauli sparsity
arXiv preprint arXiv:2510.08292, (2025).

P6. D. Stilck França, T. Möbus, C. Rouzé, A. H. Werner.
Learning and certification of local time-dependent quantum dynamics and noise
arXiv preprint arXiv:2510.08500, (2025).

P7. G. Camilo, T. O. Maciel, A. Tosta, A. Alhajri, T. L. Silva, D. Stilck França, L. Aolita.
Compilation-informed probabilistic logical-error cancellation
arXiv preprint arXiv:2508.20174, (2025).

P8. E. Onorati, C. Rouzé, D. Stilck França, J. D. Watson.
Provably efficient learning of phases of matter via dissipative evolutions
arXiv preprint arXiv:2311.07506, (2023).

P9. C. Rouzé, D. Stilck França.
Efficient learning of the structure and parameters of local Pauli noise channels
arXiv preprint arXiv:2307.02959, (2023).

P10. A. Capel, C. Rouzé, D. Stilck França.
The modified logarithmic Sobolev inequality for quantum spin systems: classical and commuting nearest neighbour interactions
arXiv preprint arXiv:2009.11817, (2020).

Invited Talks and Seminars

1. 2026-06 *Information-theoretic generalization bounds for learning from quantum data*, Scandinavian Conference on AI 2026, **Odense**, Invited keynote talk, Odense, Denmark.
2. 2026-04 *Learning and certification of local time-dependent quantum dynamics and noise*, Quantum Learning Workshop, **EPFL**, Invited talk, Lausanne, Switzerland.
3. 2026-03 *Efficient thermalization and universal quantum computing with quantum Gibbs samplers*, Quantum Resources Conference, **Tokyo**, Invited talk, Tokyo, Japan.
4. 2026-03 *Efficient thermalization and universal quantum computing with quantum Gibbs samplers*, Quantum Dunes Workshop, **Natal**, Invited talk, Natal, Brazil.
5. 2026-01 *Learning and certification of local time-dependent quantum dynamics and noise*, Workshop on Open Quantum Systems, **IPAM**, Invited talk, Los Angeles, United States.
6. 2025-12 *Efficient Hamiltonian, structure and trace distance learning of Gaussian states*, Quantum Learning Seminar, **Centre for Quantum Technologies**, Invited talk, Singapore, Singapore.
7. 2025-12 *Quantum speed-ups for solving semidefinite relaxations of polynomial optimization*, IBM Quantum Seminar, **IBM Zurich**, Invited talk, Zurich, Switzerland.
8. 2025-11 *Bounded-error quantum simulation via Hamiltonian and Lindbladian learning*, Nordic Quantum Meeting, **Nordita**, Invited talk, Stockholm, Sweden.
9. 2025-09 *Learning and certification of local time-dependent quantum dynamics and noise*, Entangle This Workshop, **Madrid**, Invited talk, Madrid, Spain.
10. 2025-09 *Efficient Hamiltonian, structure and trace distance learning of Gaussian states*, Invited Colloquium, Faculty of Physics, **Unicamp**, Invited talk, Campinas, Brazil.
11. 2025-06 *Information-theoretic generalization bounds for learning from quantum data*, Quantum Machine Learning Conference, **Galileo Galilei Institute**, Invited talk, Florence, Italy.
12. 2025-06 *Average contraction coefficients of quantum channels*, Quantum Workshop, **Aachen University**, Invited talk, Aachen, Germany.
13. 2025-06 *Quantum speed-ups for solving semidefinite relaxations of polynomial optimization*, Munich Quantum Center Meeting, **Munich Quantum Center**, Invited talk, Munich, Germany.
14. 2025-05 *Efficient thermalization and universal quantum computing with quantum Gibbs samplers*, Quantum Matter Conference, **University of Grenoble**, Invited talk, Grenoble, France.
15. 2025-05 *Efficient thermalization and universal quantum computing with quantum Gibbs samplers*, Computer Science Seminar, **Harvard University**, Invited talk, Cambridge, United States.
16. 2025-04 *Efficient thermalization and universal quantum computing with quantum Gibbs samplers*, Quantum Optimal Transport Workshop, **IPAM**, Invited talk, Los Angeles, United States.
17. 2024-11 *Learning dissipative phases of matter*, Quantum Series Seminar, **Swiss Federal Technology Institute of Lausanne**, Invited talk, Lausanne, Switzerland.
18. 2024-04 *Tutorial on Quantum Gibbs Sampling*, Simons Institute, **UC Berkeley**, Invited tutorial, Berkeley, United States.
19. 2023-12 *Learning quantum many-body states*, International Workshop on Frontiers in Quantum Information, **Haikou**, Invited talk, Haikou, China.
20. 2023-10 *Efficient learning of ground and thermal states within phases of matter*, Quantum Information Seminar, **University of the Basque Country**, Invited talk, Bilbao, Spain.
21. 2023-10 *Efficient learning of ground and thermal states within phases of matter*, QuantAlps Days, **University of Grenoble**, Invited talk, Grenoble, France.
22. 2023-08 *Efficient learning of ground and thermal states within phases of matter*, Chinese Physical Society Fall Meeting, **Ningxia University**, Invited talk, Ningxia, China.
23. 2023-07 *Efficient learning of ground and thermal states within phases of matter*, Quantum computing seminar, **University of Edinburgh**, Invited talk, Edinburgh, Scotland.
24. 2023-07 *Limitations of noisy quantum circuits*, TQC 2023, **Aveiro University**, Invited talk, Aveiro, Portugal.
25. 2023-07 *Quantum algorithm for ground state energy estimation using circuit depth with exponentially improved dependence on precision*, PsiQuantum Algorithms Seminar, **PsiQuantum**, Invited talk, Palo Alto, United States.
26. 2023-06 *Quantum concentration inequalities*, ILAS 2023, **ILAS**, Invited speaker, Madrid, Spain.
27. 2023-05 *Quantum concentration inequalities*, Quantum Trajectories Workshop, **Université Paul Sabatier**, Invited speaker, Toulouse, France.
28. 2023-03 *Limitations imposed by noise on near-term quantum algorithms for optimization*, QRC Seminar series, **Technology Innovation Institute**, Invited talk, Abu Dhabi, United Arab Emirates.

29. 2022-12 *Exponentially tighter bounds on limitations of quantum error mitigation*, Quantum computing seminar, **University of Edinburgh**, Invited talk, Edinburgh, Scotland.
30. 2022-11 *Concentration properties of shallow quantum circuits and applications to variational quantum algorithms*, Journées Informatique Quantique 2022, **Sorbonne University**, Invited talk, Paris, France.
31. 2022-09 *Mixing properties of tensor network states*, AGATES kickoff workshop, **Banach Center**, Invited talk, Warsaw, Poland.
32. 2022-08 *Limitations of noisy quantum algorithms*, Gemini Quantum Center Seminar, **University of Oslo**, Invited talk, Oslo, Norway.
33. 2022-06 *Limitations of noisy quantum annealers*, Adiabatic Quantum Computing Conference 2022, **ICTP**, Invited talk, Trieste, Italy.
34. 2022-06 *Tomography of many-body quantum states from a few copies through optimal transport*, Quantum Information Theory and Mathematical Physics 2022, **Budapest University of Technology and Economics**, Invited talk, Budapest, Hungary.
35. 2022-06 *Efficient and robust estimation of many-qubit Hamiltonians*, Control Club, **Forschungszentrum Jülich**, Invited talk, Jülich, Germany.
36. 2022-05 *Limitations of variational quantum algorithms: a quantum optimal transport approach*, Workshop Quantum Information and the Frontiers of Quantum Theory, **ENS Lyon**, Invited talk, Lyon, France.
37. 2021-10 *Limitations of optimization on noisy quantum devices*, INFORMS Annual Meeting 2021, **INFORMS**, Invited talk, Anaheim, United States.
38. 2021-09 *Limitations of optimization on noisy quantum devices*, Beyond IID in Information Theory 9, **National Taiwan University**, Invited talk, Taipei, Taiwan.
39. 2021-09 *A game of quantum advantage: linking verification and simulation*, Second Kyoto Workshop on Quantum Information, Computation, and Foundation, **Yukawa Institute for Theoretical Physics**, Invited talk, Kyoto, Japan.
40. 2021-06 *Eventually entanglement breaking Markovian dynamics: Structure and characteristic times*, Munich-Toulouse Workshop, **University of Toulouse**, Invited talk, Toulouse, France.
41. 2021-05 *Learning quantum many-body systems from a few copies*, Seminar talk, **ENS Lyon**, Invited talk, Lyon, France.
42. 2021-05 *A game of quantum advantage: linking verification and simulation*, Seminar talk, **Heinrich Heine University Duesseldorf**, Invited talk, Duesseldorf, Germany.
43. 2021-03 *Limitations of optimization algorithms on noisy quantum devices*, Seminar talk, **Sorbonne University**, Invited talk, Paris, France.
44. 2021-02 *Limitations of optimization algorithms on noisy quantum devices*, QuICS Seminar, **QUICS/University of Maryland**, Invited talk, College Park, United States.
45. 2020-10 *Fast and robust quantum state tomography from few basis measurements*, Quantum Information Theory Seminar, **University College London**, Invited talk, London, United Kingdom.
46. 2020-04 *Faster quantum and classical SDP approximations for quadratic binary optimization*, Quantum Information Seminar, **Technical University of Munich**, Invited talk, Munich, Germany.
47. 2019-10 *Faster quantum and classical SDP approximations for quadratic binary optimization*, Quantum innovators workshop, **University of Waterloo**, Invited talk, Waterloo, Canada.
48. 2019-05 *Weak simulation and benchmarking of sparse quantum circuits*, IQIM Seminar, **California Institute of Technology**, Invited talk, Pasadena, United States.
49. 2019-05 *Noise-robust exploration of quantum matter on near-term quantum devices*, Seminar talk, **Niels Bohr Institute**, Invited talk, Copenhagen, Denmark.
50. 2018-12 *Faster quantum and classical SDP approximations for quadratic binary optimization*, Quantum Information Seminar, **Massachusetts Institute of Technology**, Invited talk, Cambridge, United States.
51. 2018-05 *Hypercontractivity of quantum dynamical semigroups*, Workshop on quantum functional inequalities, **University of Toulouse**, Invited talk, Toulouse, France.
52. 2018-05 *Approximate Randomized Benchmarking for Finite Groups*, Quantum Information Theory Seminar, **University of Cambridge**, Invited talk, Cambridge, United Kingdom.
53. 2017-10 *Dimensionality reduction of SDPs through sketching*, Workshop on Probabilistic Techniques and Quantum Information Theory, **Institut Henri Poincaré**, Invited talk, Paris, France.
54. 2016-05 *Hypercontractivity and convergence in Rényi divergences for quantum semigroups*, First Meeting of the MISTEQ Project, **University of Toulouse**, Invited talk, Toulouse, France.

Contributed Conference Presentations

An asterisk indicates a presentation delivered by a co-author.

1. 2026-08 *Learning and certification of local time-dependent quantum dynamics and noise*, TQC 2026 (August 31–September 4, 2026), **Delta Hotels by Marriott Sherbrooke Conference Center**, Contributed talk, Sherbrooke, Quebec, Canada.
2. 2025-06 *Efficient thermalization and universal quantum computing with quantum Gibbs samplers*, STOC 2025, **Prague**, Contributed talk, Prague, Czech Republic.
3. *2024-09 *Information-theoretic generalization bounds for learning from quantum data*, TQC 2024, **Okinawa University**, Contributed talk, Okinawa, Japan.
4. 2024-09 *Making both ends meet: from efficient simulation to universal quantum computing with quantum Gibbs sampling*, TQC 2024, **Okinawa University**, Contributed talk, Okinawa, Japan.
5. *2024-09 *Noise-induced shallow circuits and absence of barren plateaus*, TQC 2024, **Okinawa University**, Contributed talk, Okinawa, Japan.
6. *2024-08 *Order-p quantum Wasserstein distances from couplings*, Beyond IID 2024, **University of Illinois**, Contributed talk, Urbana, United States.
7. *2024-01 *Quantum metrology in the finite-sample regime*, QIP 2024, **Taipei International Convention Center**, Contributed talk, Taipei, Taiwan.
8. *2024-01 *Efficient learning of ground and thermal states within phases of matter*, QIP 2024, **Taipei International Convention Center**, Contributed talk, Taipei, Taiwan.
9. *2024-01 *Going Beyond Gadgets: The Importance of Scalability for Analogue Quantum Simulators*, QIP 2024, **Taipei International Convention Center**, Contributed talk, Taipei, Taiwan.
10. 2023-08 *Limitations imposed by noise on near-term quantum algorithms for optimization*, Paraty Quantum, **Universidade Federal do Rio de Janeiro**, Contributed talk, Paraty, Brazil.
11. 2023-07 *Efficient learning of ground and thermal states within phases of matter*, TQC 2023, **Aveiro University**, Contributed talk, Aveiro, Portugal.
12. 2023-04 *Quantum algorithm for ground state energy estimation using circuit depth with exponentially improved dependence on precision*, QCTIP, **Riverlane**, Contributed talk, Cambridge, United Kingdom.
13. *2023-02 *Exponentially tighter bounds on error mitigation*, QIP 2023, **Ghent University**, Contributed talk, Ghent, Belgium.
14. 2023-02 *Limitations of VQAs: a quantum optimal transport approach*, QIP 2023, **Ghent University**, Contributed talk, Ghent, Belgium.
15. *2022-07 *Quantum Differential Privacy: An Information Theory Perspective*, TQC 2022, **University of Illinois Urbana-Champaign**, Contributed talk, Urbana-Champaign, United States.
16. 2022-07 *Efficient and robust estimation of many-qubit Hamiltonians*, TQC 2022, **University of Illinois Urbana-Champaign**, Contributed talk, Urbana-Champaign, United States.
17. *2022-03 *A refinement of Pinsker's inequality and applications to state tomography*, QIP 2022, **Caltech**, Contributed talk, Pasadena, United States.
18. *2021-08 *The modified logarithmic Sobolev inequality for quantum spin systems*, International Congress of Mathematical Physics 2021, **University of Geneva**, Contributed talk, Geneva, Switzerland.
19. 2021-08 *Fast and robust quantum state tomography from few basis measurements*, International Congress of Mathematical Physics 2021, **University of Geneva**, Contributed talk, Geneva, Switzerland.
20. 2021-07 *Efficient learning of quantum extensive observables*, TQC 2021, **University of Latvia**, Contributed talk, Riga, Latvia.
21. 2021-07 *Fast and robust quantum state tomography from few basis measurements*, TQC 2021, **University of Latvia**, Contributed talk, Riga, Latvia.
22. 2021-07 *A game of quantum advantage: linking verification and simulation*, TQC 2021, **University of Latvia**, Contributed talk, Riga, Latvia.
23. *2021-07 *Optimization at the boundary of the tensor network variety*, TQC 2021, **University of Latvia**, Contributed talk, Riga, Latvia.
24. 2021-02 *Limitations of optimization algorithms on noisy quantum devices*, QIP 2021, **Munich Quantum Center**, Contributed talk, Munich, Germany.

25. *2021-02 *On the entropic convergence of quantum Gibbs samplers*, QIP 2021, **Munich Quantum Center**, Contributed talk, Munich, Germany.
26. *2020-11 *The modified logarithmic Sobolev inequality for quantum spin systems: classical and commuting nearest neighbour interactions*, Beyond IID Conference, **Stanford University**, Contributed talk, Online event.
27. 2020-06 *Faster quantum and classical SDP approximations for quadratic binary optimization*, Conference on the Theory of Quantum Computation, Communication and Cryptography (TQC), **University of Latvia**, Contributed talk, Riga, Latvia.
28. *2019-07 *Group transference techniques for the estimation of the decoherence times and capacities of quantum Markov semigroups*, Beyond IID Conference, **University of Technology Sydney**, Contributed talk, Sydney, Australia.
29. *2019-07 *On entanglement breaking times for quantum Markovian evolutions*, Beyond IID Conference, **University of Technology Sydney**, Contributed talk, Sydney, Australia.
30. 2019-01 *Functional inequalities via group transference techniques and application to estimation of decoherence times and capacities*, Conference on Quantum Information Processing (QIP), **University of Colorado**, Contributed talk, Boulder, United States.
31. 2017-10 *Perfect Sampling of quantum Gibbs states*, International Conference for Quantum Information Scientists, **Max Planck Institute of Light**, Contributed talk, Erlangen, Germany.
32. 2015-06 *Logarithmic Sobolev Inequalities for Entropy Production*, 12th Central European Quantum Information Processing Workshop, **Masaryk University**, Contributed talk, Telč, Czech Republic.